METHOD STATEMENT

# Back-fill — Closed-Cell Polyethylene Foam Backer Rod for Joints

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|---|---|---|---|
| **Project Name** | | **Document No.** | OMX-MS-BACKFILL |
| **Main Contractor** | | **Revision** | 01 |
| **Consultant** | | **Date of Issue** | Nov 2026 |
| **Product** | Back-fill | **Status** | For Approval |

## 1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Back-fill on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

## 2. Scope of Work

This method statement covers the selection and installation of Back-fill closed-cell polyethylene backer rod in expansion and control joints before the application of joint sealants.

- Inspection and cleaning of joints.
- Selection of rod diameter and sealant depth.
- Installation of the backer rod.
- Inspection before sealing.
- Coordination with sealant application.

## 3. Definitions

| Term | Meaning |
|---|---|
| **Backer rod** | Foam rod placed in a joint to control sealant depth. |
| **Bond breaker** | Material that prevents the sealant bonding to the joint bottom. |
| **Three-sided adhesion** | Sealant bonded to both faces and the bottom; causes failure under movement. |
| **Width : depth ratio** | Ratio of sealant width to its depth. |

## 4. Roles & Responsibilities

| Role | Responsibility |
|---|---|
| **Project Manager** | Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor. |
| **Site Engineer** | Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records. |

| Role | Responsibility |
|---|---|
| **QA/QC Inspector** | Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist. |
| **HSE Officer** | Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise. |
| **Foreman / Applicators** | Carry out the work as per this method statement and report any deviation immediately. |
| **Storekeeper** | Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers. |

## 5. Materials

| Material | Description |
|---|---|
| **Back-fill** | Closed-cell PE backer rod, diameters to suit joint widths |
| **Joint sealant** | Mixseal PU or as specified |

### 5.1 Key product data

| Property | Typical Value |
|---|---|
| **Material** | Closed-cell polyethylene foam |
| **Density** | 35 – 40 kg/m³ |
| **Colour** | Blue / black |
| **Tensile strength** | 70 kPa |
| **Elongation at break** | 116 % |
| **Compression recovery** | > 90 % |
| **Flexibility** | down to -40 °C |
| **Thermal conductivity** | 0.043 W/m·K |

## 6. Tools & Equipment

- Blunt insertion roller or wooden tool
- Knife
- Tape measure
- Depth gauge
- Wire brush
- Blower
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

### 6.1 Personal protective equipment

- Gloves
- Safety glasses
- Safety shoes

## 7. Storage & Handling

- Store in a dry, shaded place away from direct sunlight, heat sources and sharp objects.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Store rolls and cartons flat, off the ground, protected from direct sunlight, heat, oil and mechanical damage.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

## 8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

## 9. Surface Preparation

1. Measure joint widths and confirm the movement design.
2. Clean joints of dust, laitance, old sealant, oil and loose material.
3. Ensure joint faces are dry and sound; repair damaged arrises.
4. Obtain QA/QC approval of cleaned joints before installing the rod.

## 10. Selection of Rod Diameter

1. Check the diameter and condition of the delivered rolls.
2. Select a rod diameter about 25 – 30 % larger than the joint width (compressed to about 70 % of its diameter).
3. Under water pressure, select a diameter that gives about 40 % compression.
4. Set the sealant depth: width 6 – 12 mm → depth = width; 12 – 24 mm → 12 mm; > 24 mm → half the width.

## 11. Installation

1. Unroll the rod without stretching.
2. Press the rod into the joint with a blunt roller or wooden tool to the set depth.
3. Keep a uniform depth along the joint; check with a depth gauge.

4. Butt-join ends tightly without overlapping.
5. Apply the joint sealant over the rod as soon as possible.
6. Record joint widths, rod diameters and depths.

## 12. Protection

- Protect installed rods from water, dirt and damage until sealed.
- Replace punctured or stretched rods before sealing.

## 13. Quality Control & Inspection

| Check | Stage | Acceptance Criteria | By |
|---|---|---|---|
| **Joints** | Before installation | Clean, dry, sound | QA/QC |
| **Rod diameter** | Before installation | Compressed to ~70 % in joint | Site Engineer |
| **Depth** | After installation | Uniform, as per width : depth rule | QA/QC |
| **Continuity** | After installation | No gaps; ends butt-joined | QA/QC |

### 13.1 Inspection and test plan

| Activity | Inspection / Test | Frequency | Record |
|---|---|---|---|
| **Material receipt** | Diameter, condition | Each delivery | Material inspection report |
| **Joint cleaning** | Clean, dry, sound | Each joint | Inspection request |
| **Rod installation** | Diameter, depth, continuity | Each joint | Inspection request (hold point) |
| **Sealing** | Sealant profile | Each joint | Daily report |

## 14. Defects & Remedies

| Defect | Likely Cause | Remedy |
|---|---|---|
| **Rod too loose** | Diameter too small | Replace with larger diameter |
| **Uneven sealant depth** | Rod inserted unevenly | Re-set rod to uniform depth |
| **Bubbles in sealant** | Punctured rod releasing gas | Use blunt tools only; replace punctured rod |
| **Sealant three-sided adhesion** | Rod missing or gaps | Install continuous rod |

## 15. Health, Safety & Environment

| Hazard | Control Measures |
|---|---|
| **Cuts from knives** | Cut-resistant gloves |
| **Work at height** | Approved access, harness where required |

| Hazard | Control Measures |
|---|---|
| **Fire (combustible foam)** | Keep away from heat and flames |
| **Slips and trips** | Housekeeping |

- Wear gloves when handling and cutting.
- Keep away from heat sources and open flames (combustible material).
- Follow site safety rules for work at height.
- Eyes: rinse with clean water if particles enter the eye.
- Cuts: clean and dress the wound.

### 15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

## 16. Cleaning

- No cleaning required; collect off-cuts in waste bins.

## 17. Limitations

- Do not puncture, twist or stretch the rod.
- Do not install in wet joints.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

## Appendix A — Inspection Checklist

| Inspection Item | Result | Remarks |
|---|---|---|
| **Joint widths measured** | ☐ Yes ☐ No | |
| **Joints cleaned and dry** | ☐ Yes ☐ No | |
| **Rod diameter selected** | ☐ Yes ☐ No | |
| **Rod inserted with blunt tool** | ☐ Yes ☐ No | |
| **Uniform depth verified** | ☐ Yes ☐ No | |
| **Ends butt-joined** | ☐ Yes ☐ No | |
| **No punctures** | ☐ Yes ☐ No | |
| **Sealant applied promptly** | ☐ Yes ☐ No | |

| Inspected by | Signature | Date |
|---|---|---|
| | | |